

Zachery Francis

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U.S. Citizen at Brooklyn, NY

Education & Certificates

- B.S. Information Systems & B.S. Business Management, Brooklyn College *December 2026*
Minor: Computer Science, Brooklyn College
- CodePath: Technical Interview Prep & Applied AI Engineering Certifications

Projects

Popular Vote: Full-Stack Web Application

- Created a live classroom messaging tool that lets students ask questions and helps teachers organize them during a lecture.
- Developed a live classroom Q&A platform for hosts to collect and organize audience questions in real time using React, Node.js, PostgreSQL, Socket.io, and Google OAuth.
- Clustered participant questions by semantic theme using Gemini Flash 2.5 REST APIs, reducing duplicate questions and cutting host response overhead during live sessions.
- Validated real-time backend session stability and front-end render performance by demoing to 40 simultaneous users joined via QR code under live load.
- Coordinated full-stack feature development across a 3-person Agile team using Scrum standups and Kanban tracking, mentored by a senior Shopify engineer.
- Refactored front-end components and relational database logic into a unified role-based architecture after senior engineer feedback, improving maintainability.

Calisigh: Desktop Calendar Application

- Built a private calendar application that uses artificial intelligence to help users organize their personal schedules locally.
- Developed a privacy-first desktop application for local AI-assisted scheduling using React front-end frameworks alongside Tauri and Rust backend logic.
- Embedded an Ollama 3.2 local LLM as a context-aware scheduling assistant to deliver full-stack AI features without sending calendar data to external REST APIs.
- Implemented custom alarm sounds, background themes, font selection, event duplication, and a toggleable federal, cultural, and religious holiday library.

Sudoku: Full-Stack Puzzle Web Application

- Created an online puzzle application that generates boards and allows users to solve them automatically using computer code.
- Developed a browser-based Sudoku platform for puzzle generation, validation, and algorithmic solving using React, Vite, Python, and Flask backend architecture.
- Built a photo-to-puzzle data pipeline using Gemini Flash 2.5 and PIL so users can photograph a physical board and load it directly into the full-stack solver.

Work History

Teacher's Assistant at Brooklyn College CS Dept., Brooklyn, NY

February 2026 to Present

- Assisted a university computer science classroom by grading assignments and providing technical coding help to students.
- Evaluated and debugged Java backend assignments for 60+ students using TuringsCraft CodeLab; adjusted feedback strategy based on grade outcomes until student understanding improved.
- Tracked recurring coding errors across submissions and coordinated with instructors to adjust lecture emphasis, reducing compounding mistakes across the class.
- Provided depth-level debugging feedback that drew students back voluntarily, improving understanding of core object-oriented software engineering principles.